

A Modular Classification of Pre-Descent Resistance in Accelerated Odd Collatz Dynamics

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Abstract

This paper studies finite pre-descent behavior for the accelerated odd Collatz map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

where n is a positive odd integer and $v_2(\cdot)$ denotes the exponent of 2 in the prime factorization. For an odd integer $n > 1$, the first-descent time is defined by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\},$$

whenever such an r exists. When $\tau(n)$ exists, the maximum pre-descent value is also defined by

$$M(n) = \max\{n, T(n), T^2(n), \dots, T^{\tau(n)}(n)\}$$

and the relative peak ratio by

$$\rho(n) = \frac{M(n)}{n}.$$

The pair

$$\mathcal{P}(n) = (\tau(n), \rho(n))$$

is used as a pre-descent resistance profile, separating temporal delay from temporary amplification.

The main structural contribution is a modular valuation mechanism for delayed first descent. If $m \geq 2$ and

$$n \equiv -1 \pmod{2^m},$$

then

$$v_2(3n+1) = 1.$$

More strongly, if $n = 2^m q - 1$, then

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m-1).$$

Hence

$$T^j(n) > n \quad (1 \leq j \leq m-1),$$

and, whenever $\tau(n)$ exists,

$$\tau(n) \geq m.$$

Thus the nested residue classes $n \equiv -1 \pmod{2^m}$ form a modular resistance spine with a deterministic lower bound on first-descent time.

An exact-integer computation was performed for all odd integers $3 \leq n \leq 10^7$, with a limit of 5000 accelerated odd steps. All 4,999,999 tested cases reached first descent within this limit. The

observed resistance distribution is long-tailed; temporal and amplitude resistance are distinct; and average first-descent time increases along the modular resistance spine for $m = 6, \dots, 15$ in the tested range. These computations do not prove the Collatz conjecture. They provide a reproducible finite classification of modular pre-descent resistance in accelerated odd Collatz dynamics.

Keywords: Collatz dynamics; accelerated odd Collatz map; $3x + 1$ problem; first-descent time; modular valuation; pre-descent resistance; modular resistance spine; exact integer computation; residue classes

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1 Introduction

The Collatz conjecture, also known as the $3x + 1$ problem, is a classical open problem in elementary number theory and discrete dynamical systems. Given a positive integer n , the standard Collatz map sends n to $n/2$ if n is even and to $3n + 1$ if n is odd. The conjecture asserts that repeated iteration of this rule eventually reaches 1 for every positive integer. Despite the elementary form of the map, its global behavior remains unresolved. Extensive surveys, annotated bibliographies, and collected studies of the $3x + 1$ problem and its variants are given by Lagarias and related works [1, 2, 3, 4].

Substantial partial progress has been obtained without resolving the full conjecture. Earlier work studied stopping-time phenomena and density properties of Collatz trajectories [5, 1]. Computational verification and empirical exploration have also played an important role in the development of the subject [6]. Other approaches include computer-assisted estimates and bounds for the $3x + 1$ problem [7]. More recently, Tao proved that almost all Collatz orbits attain almost bounded values, with “almost all” interpreted in the sense of logarithmic density [8]. These results illustrate a distinction that is relevant to the present paper: asymptotic, density-based, computational, and structural results may reveal significant features of Collatz dynamics without proving convergence for every positive integer.

The present paper does not attempt to prove the Collatz conjecture. Instead, it studies a finite pre-descent question for the accelerated odd Collatz map. Rather than asking whether a trajectory eventually reaches 1, the question considered here is how an odd trajectory behaves before it first falls below its own initial value. Thus the focus is shifted from the global convergence problem

$$n \longrightarrow 1$$

to the finite pre-descent condition

$$T^r(n) < n. \tag{1}$$

For a fixed starting value, first descent below the starting value is a finite trajectory property. A universal proof of finite descent for all odd starting values would have global consequences for the Collatz problem, but the present paper proves only modular lower bounds and reports finite-range exact computations.

The accelerated odd Collatz map is defined by

$$T(n) = \frac{3n + 1}{2^{v_2(3n+1)}}, \tag{2}$$

where n is odd and $v_2(3n + 1)$ is the exponent of 2 dividing $3n + 1$. This map sends one odd integer directly to the next odd integer in the corresponding Collatz trajectory by suppressing the

intermediate even divisions. Accelerated or shortcut forms of the $3x + 1$ map are standard tools for isolating the odd-to-odd dynamics of the problem [1, 9, 4].

For an odd integer $n > 1$, the first-descent time is defined by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}, \quad (3)$$

whenever such an r exists. If $\tau(n) = 1$, the accelerated odd trajectory descends immediately. Larger values of $\tau(n)$ indicate that the trajectory remains at or above its initial value for a longer initial segment. In this paper, $\tau(n)$ is used as a measure of temporal pre-descent resistance. This statistic is related to, but distinct from, the stopping-time quantities studied in the Collatz literature [5, 1, 9].

Temporal delay alone does not fully describe pre-descent behavior. A trajectory may remain above its initial value for many accelerated odd steps without rising very far, while another may rise sharply before descending after fewer steps. Therefore, whenever $\tau(n)$ exists, the maximum pre-descent value is defined as

$$M(n) = \max\{n, T(n), T^2(n), \dots, T^{\tau(n)}(n)\}, \quad (4)$$

and the relative peak ratio is defined as

$$\rho(n) = \frac{M(n)}{n}. \quad (5)$$

The pair

$$\mathcal{P}(n) = (\tau(n), \rho(n)) \quad (6)$$

is used as a two-dimensional pre-descent resistance profile, where $\tau(n)$ measures temporal resistance and $\rho(n)$ measures amplitude resistance.

The main contribution of this paper is to combine these pre-descent quantities with a modular valuation analysis. The central structural object is the nested family of residue classes

$$n \equiv -1 \pmod{2^m}. \quad (7)$$

For every $m \geq 2$, this congruence forces

$$v_2(3n + 1) = 1, \quad (8)$$

and therefore forces the first accelerated odd step to be expansionary:

$$T(n) = \frac{3n + 1}{2} > n. \quad (9)$$

More strongly, if $n = 2^m q - 1$, then the first iterates satisfy

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m - 1). \quad (10)$$

It follows that

$$T^j(n) > n \quad (1 \leq j \leq m - 1), \quad (11)$$

and hence, whenever $\tau(n)$ exists,

$$\tau(n) \geq m. \quad (12)$$

Thus the classes $n \equiv -1 \pmod{2^m}$ form a modular resistance spine with a deterministic lower bound on first-descent time. This spine is introduced as a modular classification object for identifying residue classes structurally excluded from early descent, not as a convergence theorem.

The computational component evaluates pre-descent resistance over increasing finite ranges up to 10^7 , using exact integer arithmetic for the accelerated odd iteration. Large-scale computation has long been used in the study of the $3x + 1$ problem, especially for empirical verification and structural exploration [6, 4]. In the largest run, all odd integers in the range $3 \leq n \leq 10^7$ were tested with a maximum limit of 5000 accelerated odd steps. All 4,999,999 tested cases reached first descent within this limit. The results show that the resistance distribution is long-tailed, that temporal and amplitude resistance are distinct, and that average first-descent time increases along the modular resistance spine within the tested range.

The results should be interpreted as a reproducible classification of finite pre-descent behavior in accelerated odd Collatz dynamics. The main contributions are as follows:

1. the definition of a two-dimensional pre-descent resistance profile

$$\mathcal{P}(n) = (\tau(n), \rho(n));$$

2. a modular valuation and persistence result for the classes

$$n \equiv -1 \pmod{2^m},$$

yielding the deterministic lower bound

$$\tau(n) \geq m$$

whenever $\tau(n)$ exists;

3. an exact-integer computational framework for measuring first-descent time, relative peak growth, resistance classes, and modular residue-class statistics over finite ranges;
4. a large-scale finite classification up to 10^7 , showing long-tailed resistance, separation between temporal and amplitude resistance, and concentration of temporal pre-descent resistance along the modular resistance spine.

The remainder of the paper is organized as follows. The next section defines the accelerated odd map and the pre-descent quantities. The following section develops the modular valuation structure and the deterministic lower bound underlying the resistance spine. The computational method and exact streaming implementation are then described, followed by the numerical results and their interpretation.

2 The Accelerated Odd Map and Pre-Descent Quantities

This section defines the accelerated odd Collatz map and the finite pre-descent quantities used throughout the paper. The aim is not to study terminal convergence to 1, but to measure the initial segment of an odd trajectory before it first falls below its starting value.

2.1 The accelerated odd Collatz map

Let

$$\mathbb{N}^+ = \{1, 2, 3, \dots\}$$

denote the set of positive integers, and let

$$\mathcal{O}^+ = \{1, 3, 5, 7, \dots\}$$

denote the set of positive odd integers.

The standard Collatz map is

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n+1, & n \equiv 1 \pmod{2}. \end{cases} \quad (13)$$

The Collatz conjecture asserts that repeated iteration of C eventually reaches 1 for every $n \in \mathbb{N}^+$. Background on the $3x+1$ problem, its variants, and related stopping-time questions is given in the surveys and bibliographies of Lagarias [1, 2, 3].

Definition 2.1 (Accelerated odd Collatz map). For $n \in \mathcal{O}^+$, let $v_2(3n+1)$ denote the largest integer $a \geq 1$ such that

$$2^a \mid (3n+1).$$

The accelerated odd Collatz map is defined by

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}. \quad (14)$$

Since $3n+1$ is even for every odd n , the exponent $v_2(3n+1)$ is well-defined and at least 1. Division by the full power of 2 in $3n+1$ leaves an odd integer. Hence

$$T : \mathcal{O}^+ \rightarrow \mathcal{O}^+.$$

The map T records the odd subsequence of the Collatz orbit by suppressing the intermediate even divisions. Accelerated or shortcut formulations are standard in the study of the $3x+1$ problem because they isolate the odd-to-odd dynamics [1, 9].

Remark 2.2. The fixed point 1 satisfies $T(1) = 1$. For this reason, the classification considered in this paper is restricted to odd integers $n \geq 3$.

2.2 First-descent time

The main finite trajectory statistic considered here is the first time at which the accelerated odd orbit falls below its initial value.

Definition 2.3 (First-descent time). Let $n \in \mathcal{O}^+$ with $n > 1$. If there exists an integer $r \geq 1$ such that

$$T^r(n) < n,$$

then the first-descent time of n is defined by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}. \quad (15)$$

If no such r exists, then $\tau(n)$ is left undefined.

Thus $\tau(n) = 1$ precisely when $T(n) < n$. Larger values of $\tau(n)$ indicate that the accelerated odd trajectory remains at or above its initial value for a longer initial segment.

Remark 2.4. The first-descent time is distinct from the stopping time and total stopping time usually considered in the Collatz literature. It records only the first point at which the accelerated odd trajectory becomes smaller than its starting value. Thus it is a finite pre-descent statistic rather than a terminal convergence statistic.

In a finite computation, a step limit L is imposed. If no $1 \leq r \leq L$ satisfies $T^r(n) < n$, the case is recorded as unresolved within the computational limit. This status is purely computational and does not imply the existence of a non-descending trajectory.

2.3 Resistance classes

The first-descent time induces a classification of those odd integers for which finite first descent exists.

Definition 2.5 (Resistance classes). For each integer $k \geq 1$, define

$$\mathcal{R}_k = \{n \in \mathcal{O}^+ : n > 1, \tau(n) \text{ exists, and } \tau(n) = k\}. \quad (16)$$

The class $\mathcal{R}_k$ consists of odd integers whose accelerated odd trajectories first fall below their initial value after exactly k accelerated odd steps.

The class $\mathcal{R}_1$ contains the immediate-descent cases:

$$\mathcal{R}_1 = \{n \in \mathcal{O}^+ : n > 1, T(n) < n\}.$$

The classes $\mathcal{R}_2, \mathcal{R}_3, \mathcal{R}_4, \dots$ contain increasingly delayed first-descent cases. In this sense, higher values of k represent stronger temporal pre-descent resistance.

Remark 2.6. The definition of $\mathcal{R}_k$ does not assert that every odd integer belongs to some resistance class. It classifies only those integers for which finite first descent exists. In a finite computation, each tested integer is assigned either to a class $\mathcal{R}_k$, if first descent is detected, or to an unresolved category, if no first descent is found within the prescribed step limit.

2.4 Peak growth and resistance profiles

First-descent time measures the duration of pre-descent resistance, but it does not measure the magnitude of temporary growth before descent. This motivates the following two quantities.

Definition 2.7 (Maximum pre-descent value). Let $n \in \mathcal{O}^+$ with $n > 1$, and suppose that $\tau(n)$ exists. The maximum pre-descent value is

$$M(n) = \max\{n, T(n), T^2(n), \dots, T^{\tau(n)}(n)\}. \quad (17)$$

Thus $M(n)$ records the largest odd integer reached from the starting point up to and including the first-descent step. If the trajectory descends immediately, then $M(n) = n$. If it first rises and later descends, then $M(n) > n$.

Definition 2.8 (Relative peak ratio). Let $n \in \mathcal{O}^+$ with $n > 1$, and suppose that $\tau(n)$ exists. The relative peak ratio is

$$\rho(n) = \frac{M(n)}{n}. \quad (18)$$

The ratio $\rho(n)$ normalizes the maximum pre-descent value by the starting value. Thus $\rho(n) = 1$ means that the trajectory never rises above its starting value before first descent, while $\rho(n) > 1$ means that the trajectory undergoes temporary amplification before falling below its initial value.

Definition 2.9 (Pre-descent resistance profile). Let $n \in \mathcal{O}^+$ with $n > 1$, and suppose that $\tau(n)$ exists. The pre-descent resistance profile of n is

$$\mathcal{P}(n) = (\tau(n), \rho(n)). \quad (19)$$

The first component of $\mathcal{P}(n)$ measures temporal resistance, while the second component measures amplitude resistance. This two-dimensional profile is used because temporal resistance and amplitude resistance need not coincide. The numerical results below show that the largest observed first-descent time and the largest observed relative peak ratio occur at different starting values in the tested range.

2.5 Interpretation of pre-descent resistance

The pre-descent resistance framework is a classification framework, not a convergence argument. It does not attempt to prove that all Collatz trajectories eventually reach 1. Instead, it measures the finite initial behavior of an accelerated odd trajectory before it first falls below its starting value.

The distinction is between the global Collatz question,

$$\text{Does every } n \in \mathbb{N}^+ \text{ eventually reach 1?}$$

and the finite pre-descent question,

$$\text{For a given odd } n > 1, \text{ when does } T^r(n) < n \text{ first occur?}$$

The second question can be studied by finite computation and modular classification without resolving the first for all positive integers.

The quantities introduced above provide a finite trajectory vocabulary: $\tau(n)$ measures first-descent delay; $\mathcal{R}_k$ groups integers by temporal resistance class; $M(n)$ records the largest odd value reached before first descent; $\rho(n)$ normalizes this peak by the initial value; and $\mathcal{P}(n)$ combines temporal and amplitude resistance into a two-dimensional profile. This vocabulary allows individual integers, residue classes, and modular refinement chains to be compared according to their pre-descent behavior.

3 Modular Valuation Structure and the Resistance Spine

This section develops the modular structure underlying the principal resistance family studied in the computations. The key observation is that certain residue classes modulo powers of 2 force minimal division in the accelerated odd map, producing a deterministic lower bound for first-descent time.

3.1 Residue classes modulo powers of 2

The accelerated odd map depends on the exponent

$$v_2(3n + 1),$$

which measures divisibility by powers of 2. It is therefore natural to group odd integers by residue classes modulo 2^m :

$$n \equiv a \pmod{2^m}.$$

Such residue classes determine congruence information about $3n + 1$ and hence influence the first odd-to-odd step under T .

The computations reported below show that the class

$$n \equiv 63 \pmod{64}$$

has the largest average first-descent time among residue classes modulo 64 in the tested range. Refining this class modulo higher powers of 2 leads to the nested chain

$$63 \pmod{64}, \quad 127 \pmod{128}, \quad 255 \pmod{256}, \quad 511 \pmod{512}, \quad 1023 \pmod{1024}, \dots$$

This chain is represented generally by

$$2^m - 1 \pmod{2^m},$$

or equivalently by

$$n \equiv -1 \pmod{2^m}.$$

This motivates the following definition.

3.2 The modular resistance spine

Definition 3.1 (Modular resistance spine). For $m \geq 2$, define the modular resistance spine at level m by

$$\mathcal{S}_m = \{n \in \mathcal{O}^+ : n \equiv -1 \pmod{2^m}\}. \quad (20)$$

Equivalently,

$$\mathcal{S}_m = \{n \in \mathcal{O}^+ : n \equiv 2^m - 1 \pmod{2^m}\}.$$

The term “spine” refers to the nesting property

$$\mathcal{S}_{m+1} \subseteq \mathcal{S}_m.$$

Indeed, if

$$n \equiv -1 \pmod{2^{m+1}},$$

then necessarily

$$n \equiv -1 \pmod{2^m}.$$

Thus each successive level of the spine is a refinement of the previous one.

Remark 3.2. The term “spine” is used only as a classification term for the nested family $\{\mathcal{S}_m\}_{m \geq 2}$. It does not imply non-convergence, infinite resistance, or any obstruction to the Collatz conjecture.

3.3 Valuation and first-step expansion on the spine

The modular resistance spine has a simple valuation-theoretic basis: its defining congruence forces the smallest possible value of $v_2(3n + 1)$ at the first accelerated odd step.

Lemma 3.3 (Valuation on the modular resistance spine). *Let $m \geq 2$. If*

$$n \equiv -1 \pmod{2^m},$$

then

$$v_2(3n + 1) = 1. \quad (21)$$

Proof. Assume that $n \equiv -1 \pmod{2^m}$. Then there exists an integer $q \geq 1$ such that

$$n = 2^m q - 1.$$

Substitution gives

$$3n + 1 = 3(2^m q - 1) + 1 = 3 \cdot 2^m q - 2.$$

Factoring out 2 yields

$$3n + 1 = 2(3 \cdot 2^{m-1} q - 1).$$

Since $m \geq 2$, the integer $3 \cdot 2^{m-1} q$ is even. Therefore

$$3 \cdot 2^{m-1} q - 1$$

is odd. Thus $3n + 1$ is divisible by 2 but not by 4, and hence

$$v_2(3n + 1) = 1.$$

□

Proposition 3.4 (First-step expansion on the spine). *Let $m \geq 2$. If $n \in \mathcal{S}_m$, then*

$$T(n) = \frac{3n+1}{2} > n. \quad (22)$$

Consequently, $n \notin \mathcal{R}_1$. If $\tau(n)$ exists, then $\tau(n) > 1$.

Proof. By Lemma 3.3, every $n \in \mathcal{S}_m$ satisfies

$$v_2(3n+1) = 1.$$

Therefore

$$T(n) = \frac{3n+1}{2}.$$

The inequality

$$\frac{3n+1}{2} > n$$

is equivalent to $n+1 > 0$, which holds for every positive integer n . Hence $T(n) > n$, so n is not an immediate-descent case. Thus $n \notin \mathcal{R}_1$, and if $\tau(n)$ exists, then $\tau(n) > 1$. $\square$

Remark 3.5. Proposition 3.4 does not imply that elements of the modular resistance spine fail to descend, nor does it imply any obstruction to the Collatz conjecture. It only shows that these elements are structurally excluded from immediate descent under the accelerated odd map.

3.4 Persistence of the spine under accelerated iteration

The preceding result gives a first-step obstruction to immediate descent. A stronger property is that the congruence

$$n \equiv -1 \pmod{2^m}$$

persists under accelerated odd iteration in a weakened form. Each accelerated step reduces the modular depth by one while preserving the same structural form.

Lemma 3.6 (Persistence of the modular spine). *Let $m \geq 2$, and suppose that*

$$n = 2^m q - 1$$

for some integer $q \geq 1$. Then, for every integer j satisfying

$$0 \leq j \leq m-1,$$

one has

$$T^j(n) = 3^j 2^{m-j} q - 1. \quad (23)$$

In particular,

$$T^j(n) \equiv -1 \pmod{2^{m-j}} \quad (0 \leq j \leq m-1). \quad (24)$$

Proof. The formula is proved by induction on j . For $j = 0$,

$$T^0(n) = n = 2^m q - 1,$$

which agrees with (23).

Assume that (23) holds for some j with

$$0 \leq j \leq m-2.$$

Then

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

Hence

$$3T^j(n) + 1 = 3(3^j 2^{m-j} q - 1) + 1 = 3^{j+1} 2^{m-j} q - 2.$$

Factoring out 2 gives

$$3T^j(n) + 1 = 2(3^{j+1} 2^{m-j-1} q - 1).$$

Since $j \leq m - 2$, one has $m - j - 1 \geq 1$. Hence

$$3^{j+1} 2^{m-j-1} q$$

is even, and therefore

$$3^{j+1} 2^{m-j-1} q - 1$$

is odd. Thus

$$v_2(3T^j(n) + 1) = 1.$$

Consequently,

$$T^{j+1}(n) = \frac{3T^j(n) + 1}{2} = 3^{j+1} 2^{m-j-1} q - 1.$$

This proves the induction step, and therefore (23) holds for all $0 \leq j \leq m - 1$.

The congruence statement follows from

$$T^j(n) + 1 = 3^j 2^{m-j} q,$$

which is divisible by 2^{m-j} . Hence

$$T^j(n) \equiv -1 \pmod{2^{m-j}}.$$

□

Remark 3.7. Lemma 3.6 shows that the modular resistance spine is not merely a first-step phenomenon. Starting from depth m , the orbit remains on a lower-depth spine for the first $m - 1$ accelerated odd iterates.

3.5 A lower bound for first-descent time on the spine

The persistence property implies a deterministic lower bound for first-descent time. Thus elements of $\mathcal{S}_m$ are not only excluded from immediate descent; they are excluded from descent for a finite number of accelerated odd steps controlled by the modular depth m .

Proposition 3.8 (Lower bound for first-descent time on the spine). *Let $m \geq 2$. If*

$$n \in \mathcal{S}_m,$$

then

$$T^j(n) > n \quad \text{for every } 1 \leq j \leq m - 1. \tag{25}$$

Consequently, if $\tau(n)$ exists, then

$$\tau(n) \geq m. \tag{26}$$

Proof. Since $n \in \mathcal{S}_m$, there exists an integer $q \geq 1$ such that

$$n = 2^m q - 1.$$

By Lemma 3.6,

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m-1).$$

For $1 \leq j \leq m-1$, the inequality

$$T^j(n) > n$$

is equivalent to

$$3^j 2^{m-j} q - 1 > 2^m q - 1.$$

After cancellation and division by $2^{m-j} q > 0$, this becomes

$$3^j > 2^j,$$

which holds for every $j \geq 1$. Hence

$$T^j(n) > n \quad (1 \leq j \leq m-1).$$

By definition,

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}.$$

Since no iterate $T^j(n)$ with $1 \leq j \leq m-1$ is below n , the first possible descent can occur no earlier than the m -th accelerated odd step. Therefore, if $\tau(n)$ exists, then

$$\tau(n) \geq m.$$

□

Remark 3.9. Proposition 3.8 is a finite pre-descent result, not a convergence theorem. It gives a guaranteed lower bound on the first-descent time for spine elements, but it does not determine whether descent occurs after the m -th accelerated odd step.

3.6 Division deficiency and classification role

The preceding propositions provide an algebraic explanation for why the spine is a natural high-resistance family. In the accelerated odd map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

the numerator $3n+1$ contributes expansion, while the denominator $2^{v_2(3n+1)}$ contributes contraction. When $v_2(3n+1) = 1$, the accelerated step divides only by 2, which is insufficient to offset the multiplication by 3. Thus the accelerated odd step is forced upward.

The phrase *division deficiency* is used descriptively for this situation: the power of 2 removed after the odd step is too small to compensate for the expansion introduced by $3n+1$. The modular resistance spine represents a canonical family of division-deficient classes. More strongly, Lemma 3.6 shows that if

$$n = 2^m q - 1,$$

then the first $m-1$ accelerated odd iterates satisfy

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m-1).$$

Thus each early iterate remains congruent to -1 modulo a decreasing power of 2. In this sense, the spine does not merely force one upward step; it forces a finite sequence of upward accelerated odd steps whose length is controlled by the modular depth m .

The spine is therefore used as a classification object rather than as a convergence theorem. Its strengthened defining implication is

$$n \equiv -1 \pmod{2^m} \implies \tau(n) \geq m, \quad (27)$$

whenever $\tau(n)$ exists. Thus every spine element has built-in finite pre-descent resistance. The computations reported below examine whether this deterministic lower bound is associated with longer average first-descent time under modular refinement.

3.7 Relationship to resistance profiles

The modular resistance spine interacts naturally with the resistance profile

$$\mathcal{P}(n) = (\tau(n), \rho(n)).$$

For a finite computational bound N , define the truncated spine class

$$\mathcal{S}_m(N) = \{n \in \mathcal{S}_m : 3 \leq n \leq N\}. \quad (28)$$

For each such class, aggregate resistance statistics may be computed, including

$$\bar{\tau}(\mathcal{S}_m(N)), \quad \max_{n \in \mathcal{S}_m(N)} \tau(n), \quad \bar{\rho}(\mathcal{S}_m(N)), \quad \max_{n \in \mathcal{S}_m(N)} \rho(n).$$

These quantities allow finite spine classes to be compared across increasing values of m .

The lower bound

$$\tau(n) \geq m$$

shows that increasing modular depth imposes a stronger deterministic pre-descent constraint on individual spine elements. The computational averages then measure how much additional delay occurs beyond this guaranteed minimum. Thus, for finite N , the quantity

$$E_m(N) = \bar{\tau}(\mathcal{S}_m(N)) - m \quad (29)$$

is interpreted as the average excess temporal resistance beyond the deterministic spine lower bound.

The computations reported below show that temporal resistance, measured by

$$\bar{\tau}(\mathcal{S}_m(N)),$$

increases along the spine for $m = 6, \dots, 15$ when $N = 10^7$. Amplitude resistance, measured by

$$\bar{\rho}(\mathcal{S}_m(N)),$$

is also informative, although it is more sensitive to extreme trajectory excursions and is not strictly monotone in the tested range.

Thus the modular resistance spine is used primarily as a finite classification family for temporal pre-descent resistance, while also providing a structured setting for comparing amplitude resistance across refined residue classes.

Remark 3.10. All aggregate quantities in this paper are finite-sample statistics over explicitly bounded computational ranges. No limiting distribution or asymptotic monotonicity is asserted.

4 Computational Method

This section specifies the exact-integer computational procedure used to evaluate the finite pre-descent quantities defined above. The computation was designed to classify finite pre-descent behavior in accelerated odd Collatz dynamics, not to prove the Collatz conjecture. Accordingly, the quantities tracked were first-descent time, maximum pre-descent value, relative peak ratio, resistance class, residue class, and membership in the modular resistance spine.

4.1 Computational setting

All accelerated odd Collatz iterations were performed using exact integer arithmetic in Python. This is essential because the map

$$T(n) = \frac{3n + 1}{2^{v_2(3n+1)}}$$

is an integer transformation, and no rounding is involved in the trajectory computation. Floating-point arithmetic was used only for reporting normalized ratios and aggregate averages, such as

$$\rho(n) = \frac{M(n)}{n}.$$

The computations were carried out for the upper bounds

$$N = 10^4, \quad 10^5, \quad 10^6, \quad 10^7.$$

The largest experiment examined the odd integers in the range

$$3 \leq n \leq 10,000,000,$$

which gives

$$4,999,999$$

tested odd integers after excluding the fixed point $n = 1$. For the largest experiment, the maximum search limit was set to

$$L = 5000$$

accelerated odd steps. If no first descent was found within this limit, the case was recorded as unresolved within the computational limit.

4.2 General computational procedure

For each tested odd integer n , the accelerated odd map was iterated as

$$n, T(n), T^2(n), T^3(n), \dots$$

until either a value below n was reached or the prescribed step limit L was exhausted.

If a first descent was detected within the step limit, the first-descent time was recorded as

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}.$$

During the same iteration, the maximum odd value reached up to and including the first-descent step was recorded as

$$M(n) = \max\{n, T(n), T^2(n), \dots, T^{\tau(n)}(n)\}.$$

The relative peak ratio was then computed as

$$\rho(n) = \frac{M(n)}{n}.$$

The observed resistance class was assigned as

$$\mathcal{R}_{\tau(n)}.$$

Thus the basic output for a descended case was

$$n, \quad \tau(n), \quad M(n), \quad \rho(n), \quad \mathcal{R}_{\tau(n)}.$$

If no descent was detected within L accelerated odd steps, the case was recorded as unresolved within the computational limit.

For the smaller computations, individual profiles and trajectories were retained for inspection. For the largest computation, full trajectories were not stored. Instead, a streaming procedure was used:

compute profile $\longrightarrow$ update aggregate statistics $\longrightarrow$ discard individual trajectory.

This procedure allowed aggregate statistics to be computed over millions of odd integers without storing all trajectories in memory.

4.3 Four-stage computational design

The computation was organized into four stages. Each stage increased the upper bound N , increased or preserved the step limit L , and added a specific analytical component. The staged design allowed the definitions to be checked at small scale, modular refinement to be examined at intermediate scale, and final aggregate statistics to be computed by memory-efficient streaming.

Table 1: Summary of the four computational stages.

Stage	Upper bound N	Step limit L	Main purpose
1	10^4	1000	Initial profiles and modulo 64 analysis
2	10^5	2000	Modular refinement across powers of 2
3	10^6	3000	Large-scale profiles and spine summaries
4	10^7	5000	Streaming aggregation and final statistics

In Stage 1, all odd integers $3 \leq n \leq 10^4$ were examined. For each tested integer, the quantities $\tau(n)$, $M(n)$, $\rho(n)$, and the corresponding resistance class were recorded. Full odd-only trajectories were also stored for inspection. This stage verified that the definitions were computationally operational and provided an initial residue-class analysis modulo $2^6 = 64$.

In Stage 2, the upper bound was increased to $N = 10^5$. The main purpose was to examine whether high-resistance behavior observed modulo 64 persisted under modular refinement. Residue classes were compared modulo

$$64, \quad 128, \quad 256, \quad 512, \quad 1024.$$

For each modulus, the aggregate statistics recorded were

$$\text{count}, \quad \bar{\tau}, \quad \max \tau, \quad \bar{\rho}, \quad \max \rho.$$

This stage identified the refinement chain

$$63 \pmod{64}, \quad 127 \pmod{128}, \quad 255 \pmod{256}, \quad 511 \pmod{512}, \quad 1023 \pmod{1024},$$

which is represented generally by

$$n \equiv -1 \pmod{2^m}.$$

In Stage 3, the upper bound was increased to $N = 10^6$. Full trajectories were no longer stored. Instead, only profile-level quantities needed for classification were retained:

$$n, \quad \tau(n), \quad M(n), \quad \rho(n), \quad \mathcal{R}_{\tau(n)}, \quad \text{status}.$$

This stage produced modular summaries, retained top temporal-resistance cases, retained top amplitude-resistance cases, and spine-level aggregate statistics for modular levels $m = 6, \dots, 13$.

In Stage 4, the upper bound was increased to $N = 10^7$, with step limit $L = 5000$. This stage examined

$$4,999,999$$

odd integers in the range

$$3 \leq n \leq 10,000,000.$$

Because storing all trajectories at this scale would be inefficient, the final computation used the streaming procedure described above. This final stage generated the main large-scale dataset used in the results section, including the resistance-class distribution, modular residue summaries, retained top temporal and amplitude resistance cases, and modular resistance spine statistics.

4.4 Modular aggregation

For each tested odd integer n , its residue class modulo powers of 2 was evaluated. For each modulus 2^m , the residue was computed as

$$a \equiv n \pmod{2^m}.$$

For each residue class $(2^m, a)$, the following aggregate statistics were maintained:

$$\text{count}(2^m, a), \quad \bar{\tau}(2^m, a), \quad \max \tau(2^m, a), \quad \bar{\rho}(2^m, a), \quad \max \rho(2^m, a).$$

These statistics allow residue classes to be ranked by temporal resistance and amplitude resistance.

The largest experiment used the modular levels

$$m = 6, 7, 8, 9, 10, 11, 12, 13, 14, 15.$$

Equivalently, the corresponding moduli were

$$64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768.$$

This range of moduli was used to examine how resistance statistics change under refinement by powers of 2.

4.5 Spine aggregation

In addition to general modular aggregation, the largest computation specifically tracked the modular resistance spine. For the finite upper bound N , the truncated spine class is

$$\mathcal{S}_m(N) = \{n \in \mathcal{S}_m : 3 \leq n \leq N\}.$$

Since

$$\mathcal{S}_m = \{n \in \mathcal{O}^+ : n \equiv -1 \pmod{2^m}\},$$

the corresponding residue at level m is

$$2^m - 1.$$

Thus, for each modular level m , the class

$$n \equiv 2^m - 1 \pmod{2^m}$$

was extracted within the tested range.

For each truncated spine class $\mathcal{S}_m(N)$, the following quantities were recorded:

$$\begin{aligned} \text{count}(\mathcal{S}_m(N)), \quad \bar{\tau}(\mathcal{S}_m(N)), \quad \max_{n \in \mathcal{S}_m(N)} \tau(n), \\ \bar{\rho}(\mathcal{S}_m(N)), \quad \max_{n \in \mathcal{S}_m(N)} \rho(n). \end{aligned}$$

The average excess temporal resistance

$$E_m(N) = \bar{\tau}(\mathcal{S}_m(N)) - m$$

was also computed. This quantity measures the mean delay beyond the deterministic lower bound $\tau(n) \geq m$ established for elements of the spine.

These finite-sample statistics were used to examine whether temporal pre-descent resistance increases along the refinement chain

$$n \equiv -1 \pmod{2^m}.$$

4.6 Top resistance cases

The strongest observed individual cases of temporal and amplitude resistance were also retained. Temporal resistance cases were ranked by

$$\tau(n),$$

whereas amplitude resistance cases were ranked by

$$\rho(n).$$

For the largest experiment, the top 100 cases in each category were retained using heap-based tracking.

These records were used to identify the largest observed first-descent time and the largest observed relative peak ratio within the tested range. Since these are finite computational maxima, they are reported only as observed values for the specified range

$$3 \leq n \leq 10^7.$$

4.7 Computational limits and interpretation

All computations in this paper are finite. Both the search range and the step limit are bounded. Therefore, the results are interpreted as computational evidence for the proposed classification framework, not as a proof of the Collatz conjecture.

In the largest computation, the assertion that all tested cases reached first descent means precisely that for every tested odd integer

$$3 \leq n \leq 10^7,$$

an integer r satisfying

$$1 \leq r \leq 5000$$

and

$$T^r(n) < n$$

was found. This does not imply that all positive integers reach 1, nor does it exclude behavior outside the tested range. The role of the computation is to show that pre-descent resistance can be measured, compared, and organized across large finite sets of odd integers.

5 Computational Results

This section reports the main results obtained from the largest computation. The experiment examined the odd integers in the range

$$3 \leq n \leq 10,000,000,$$

corresponding to

$$4,999,999$$

tested odd integers. The step limit was

$$L = 5000.$$

The results are organized around four finite-sample observations:

1. all tested cases reached first descent within the step limit;
2. the distribution of resistance classes is long-tailed;
3. temporal resistance and amplitude resistance are distinct;
4. resistance statistics are not uniformly distributed across residue classes, with strong concentration at $63 \pmod{64}$ and along the modular resistance spine.

5.1 Overall descent status

For each tested odd integer n , the quantities $\tau(n)$, $M(n)$, and $\rho(n)$ were evaluated. The largest experiment produced

$$\text{descended cases} = 4,999,999, \quad \text{unresolved cases} = 0.$$

Thus every tested odd integer reached first descent within the limit $L = 5000$. Equivalently, for every tested odd integer n , an integer r satisfying

$$1 \leq r \leq 5000$$

and

$$T^r(n) < n$$

was found.

This is a finite computational result. It does not establish that every positive integer reaches 1, nor does it prove that every odd integer has finite first-descent time.

5.2 Long-tailed distribution of resistance classes

The observed resistance classes are unevenly distributed. Many tested odd integers enter low resistance classes, while sparse high-resistance classes occur far into the tail.

In the 10^7 experiment, the largest observed first-descent time was

$$\tau(8,088,063) = 155.$$

For this integer,

$$M(8,088,063) = 5,385,051,557,$$

and

$$\rho(8,088,063) = 665.8024.$$

Thus,

$$8,088,063 \in \mathcal{R}_{155}.$$

This indicates that, although many tested odd integers descend relatively quickly, some trajectories remain at or above their starting values for substantially longer finite initial segments. Hence the observed resistance-class distribution is long-tailed within the tested range.

5.3 Temporal and amplitude resistance

The computation also indicates that temporal resistance and amplitude resistance are distinct finite-sample features. Temporal resistance is measured by $\tau(n)$, whereas amplitude resistance is measured by $\rho(n)$.

The largest observed temporal resistance occurred at

$$n = 8,088,063, \quad \tau(n) = 155.$$

By contrast, the largest observed amplitude resistance occurred at

$$n = 6,631,675.$$

For this integer,

$$\tau(n) = 140,$$

$$M(n) = 20,114,203,639,877,$$

and

$$\rho(n) = 3,033,050.2686.$$

Thus the largest observed first-descent time and the largest observed relative peak ratio occur at different starting values. This supports the use of the two-dimensional resistance profile

$$\mathcal{P}(n) = (\tau(n), \rho(n)),$$

because delay before descent and peak amplification measure different aspects of finite pre-descent behavior.

5.4 Modulo 64 residue-class statistics

Pre-descent resistance is not uniformly distributed across residue classes in the tested range. At the modulo 64 level, the residue class

$$n \equiv 63 \pmod{64}$$

had the largest observed average first-descent time.

For this class, the 10^7 experiment produced

$$\text{count} = 156,250,$$

$$\bar{\tau} = 15.5684,$$

$$\max \tau = 155,$$

$$\bar{\rho} = 82.3772,$$

and

$$\max \rho = 2,396,483.7836.$$

The next largest average first-descent times modulo 64 were near 10.56. For example,

$$n \equiv 27 \pmod{64}$$

had

$$\bar{\tau} = 10.5626,$$

and

$$n \equiv 31 \pmod{64}$$

had

$$\bar{\tau} = 10.5617.$$

Thus, within the tested range, the class $63 \pmod{64}$ was separated from the next highest residue classes by approximately five accelerated odd steps in average first-descent time.

This finite-sample separation motivates the modular refinement analysis below and is consistent with the valuation structure of the modular resistance spine.

5.5 Spine statistics and excess temporal resistance

The central modular finding concerns the truncated modular resistance spine

$$\mathcal{S}_m(10^7) = \{n \in \mathcal{S}_m : 3 \leq n \leq 10^7\}.$$

For $m = 6, \dots, 15$, the average first-descent time increases along this spine in the 10^7 experiment.

Because Proposition 3.8 gives the deterministic lower bound

$$\tau(n) \geq m \quad (n \in \mathcal{S}_m),$$

it is useful to compare the observed average first-descent time with this guaranteed minimum. The excess temporal resistance is therefore recorded as

$$E_m(10^7) = \bar{\tau}(\mathcal{S}_m(10^7)) - m.$$

Table 2: Average and excess first-descent time along the modular resistance spine in the 10^7 experiment.

m	Spine class	Count	$\bar{\tau}$	$E_m(10^7)$
6	63 (mod 64)	156250	15.5684	9.5684
7	127 (mod 128)	78125	17.9263	10.9263
8	255 (mod 256)	39062	20.2822	12.2822
9	511 (mod 512)	19531	22.6321	13.6321
10	1023 (mod 1024)	9765	24.9493	14.9493
11	2047 (mod 2048)	4882	27.5072	16.5072
12	4095 (mod 4096)	2441	29.8894	17.8894
13	8191 (mod 8192)	1220	32.1492	19.1492
14	16383 (mod 16384)	610	35.0459	21.0459
15	32767 (mod 32768)	305	37.3607	22.3607

The values in Table 2 show a monotone increase in average first-descent time along the finite refinement chain

$$n \equiv -1 \pmod{2^m}, \quad m = 6, \dots, 15,$$

within the tested range. The same table also shows that the average excess temporal resistance $E_m(10^7)$ remains positive and increases across the examined spine levels. Thus the observed delay is not merely the deterministic lower bound $\tau(n) \geq m$; it includes additional average delay beyond the guaranteed minimum.

This finite-sample monotonicity is consistent with the valuation and persistence results proved above. The computation therefore indicates that the modular spine is associated not only with forced early non-descent, but also with elevated average temporal pre-descent resistance under modular refinement.

5.6 Amplitude resistance along the spine

The relative peak ratio

$$\rho(n) = \frac{M(n)}{n}$$

was also examined along the same truncated spine classes.

Table 3: Average relative peak ratio along the modular resistance spine in the 10^7 experiment.

m	Spine class	Count	$\bar{\rho}$
6	63 (mod 64)	156250	82.3772
7	127 (mod 128)	78125	130.6475
8	255 (mod 256)	39062	209.8512
9	511 (mod 512)	19531	344.3314
10	1023 (mod 1024)	9765	298.3330
11	2047 (mod 2048)	4882	461.4293
12	4095 (mod 4096)	2441	605.4558
13	8191 (mod 8192)	1220	935.5016
14	16383 (mod 16384)	610	1452.2820
15	32767 (mod 32768)	305	1237.3507

Unlike $\bar{\tau}$, the average relative peak ratio is not monotone in m . For example, it decreases from

$$344.3314$$

at $m = 9$ to

$$298.3330$$

at $m = 10$, and from

$$1452.2820$$

at $m = 14$ to

$$1237.3507$$

at $m = 15$. This does not contradict the temporal spine pattern. Rather, it shows that amplitude resistance is more sensitive to extreme trajectory excursions. Since $\rho(n)$ records the largest relative peak before first descent, a small number of large excursions can substantially affect $\bar{\rho}$.

Thus, in the present computation, the modular resistance spine is more stable as a temporal-resistance structure than as a strictly monotone amplitude-resistance structure.

6 Discussion

The results support the central claim that finite pre-descent resistance in accelerated odd Collatz dynamics can be measured, classified, and compared through modular residue structure over bounded computational ranges. The framework does not resolve the Collatz conjecture. Rather, it provides a finite classification of trajectory behavior before an accelerated odd orbit first falls below its starting value.

The first-descent time $\tau(n)$ should be interpreted as a local pre-descent statistic rather than as a terminal convergence statistic. It records the first point at which the accelerated odd trajectory becomes smaller than its own starting value. Therefore, the fact that all tested odd integers below 10^7 reached first descent within the imposed step limit does not imply that all positive integers reach 1. It means only that, within the tested range, every examined accelerated odd trajectory dropped below its initial value within the selected computational limit.

The computations also show that temporal resistance and amplitude resistance are distinct finite-sample features. The largest observed first-descent time occurred at

$$n = 8,088,063, \quad \tau(n) = 155,$$

whereas the largest observed relative peak ratio occurred at

$$n = 6,631,675, \quad \rho(n) = 3,033,050.2686.$$

Thus the trajectory with the longest observed delay before first descent was not the same trajectory with the largest observed relative peak. This supports the use of the two-dimensional profile

$$\mathcal{P}(n) = (\tau(n), \rho(n)).$$

The modular resistance spine

$$\mathcal{S}_m = \{n \in \mathcal{O}^+ : n \equiv -1 \pmod{2^m}\}$$

has a direct algebraic interpretation. For $m \geq 2$, every element of this class satisfies

$$v_2(3n + 1) = 1,$$

and hence

$$T(n) = \frac{3n+1}{2} > n.$$

The persistence result shows more strongly that if $n = 2^m q - 1$, then

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m-1),$$

so that $T^j(n) > n$ for $1 \leq j \leq m-1$. Consequently, whenever $\tau(n)$ exists,

$$\tau(n) \geq m.$$

This gives a deterministic lower-bound mechanism for finite first-descent time.

The computational results suggest that this deterministic lower bound is accompanied by additional average delay in the tested range. In the 10^7 experiment, the average first-descent time along the spine increased monotonically for $m = 6, \dots, 15$. Moreover, the excess temporal resistance

$$E_m(10^7) = \bar{\tau}(\mathcal{S}_m(10^7)) - m$$

also increased over the same range. These observations remain finite-sample results. They do not establish asymptotic monotonicity of $\bar{\tau}(\mathcal{S}_m(N))$ or $E_m(N)$, nor do they imply convergence of all spine elements.

Several limitations follow. First, the computation is finite and restricted to

$$3 \leq n \leq 10^7.$$

Second, the step limit is finite. Although no unresolved case occurred within 5000 accelerated odd steps, this only describes the tested range and threshold. Third, the monotone behavior of $\bar{\tau}$ and $E_m(10^7)$ along the spine is a finite-sample observation, not an asymptotic theorem. Fourth, amplitude statistics such as $\bar{\rho}$ are sensitive to extreme trajectory excursions and should therefore be interpreted with more caution than average first-descent time. Finally, the present analysis focuses on residue classes modulo powers of 2, which are natural for the accelerated odd map, but other modular structures may also be relevant.

7 Conclusion

This paper introduced a modular classification framework for finite pre-descent resistance in accelerated odd Collatz dynamics. The objective was not to prove the Collatz conjecture, but to measure and organize how odd integers behave before they first fall below their starting value.

The framework is based on the accelerated odd map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

the first-descent time $\tau(n)$, the maximum pre-descent value $M(n)$, the relative peak ratio $\rho(n)$, and the two-dimensional resistance profile

$$\mathcal{P}(n) = (\tau(n), \rho(n)).$$

These quantities separate temporal delay from amplitude growth and provide a finite classification of observed odd trajectories.

The main theoretical contribution is the modular valuation and persistence structure of the classes

$$n \equiv -1 \pmod{2^m}.$$

For every $m \geq 2$, this congruence implies

$$v_2(3n + 1) = 1,$$

and hence

$$T(n) = \frac{3n + 1}{2} > n.$$

More strongly, if $n = 2^m q - 1$, then

$$T^j(n) = 3^j 2^{m-j} q - 1 \quad (0 \leq j \leq m - 1).$$

It follows that

$$T^j(n) > n \quad (1 \leq j \leq m - 1),$$

and consequently, whenever $\tau(n)$ exists,

$$\tau(n) \geq m.$$

Thus the modular resistance spine is structurally excluded not only from immediate descent, but from descent during the first $m - 1$ accelerated odd steps.

The computational study used exact integer arithmetic over increasing finite ranges up to 10^7 . In the largest experiment, 4,999,999 odd integers in the range

$$3 \leq n \leq 10,000,000$$

were tested with a limit of 5000 accelerated odd steps. All tested cases reached first descent within this limit. The largest observed first-descent time was

$$\tau(8,088,063) = 155,$$

while the largest observed relative peak ratio occurred at

$$n = 6,631,675, \quad \rho(n) = 3,033,050.2686.$$

These values show that temporal and amplitude resistance are distinct finite-sample features.

The modular spine displayed a monotone increase in average first-descent time for $m = 6, \dots, 15$ in the 10^7 experiment, increasing from

$$\bar{\tau} = 15.5684$$

to

$$\bar{\tau} = 37.3607.$$

After subtraction of the deterministic lower bound, the excess temporal resistance

$$E_m(10^7) = \bar{\tau}(\mathcal{S}_m(10^7)) - m$$

also increased over the same range. This supports the interpretation of the spine as a finite-sample concentration of temporal pre-descent resistance beyond its guaranteed lower bound.

These results do not constitute a proof of the Collatz conjecture. They show that finite pre-descent resistance is measurable, classifiable, and organized by modular residue structure within the tested range. Future work may extend the computational range, examine additional modular families, and test whether cumulative division-deficiency statistics such as

$$D_r(n) = \sum_{i=0}^{r-1} (\log_2 3 - v_2(3T^i(n) + 1))$$

help predict high values of $\tau(n)$ or $\rho(n)$.

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Declarations

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The author declares that no funding was received for this study.

Conflict of Interest

The author declares that there is no conflict of interest.

Data Availability

The numerical output tables generated during the computational study are available from the author upon reasonable request.

Code Availability

The computations were implemented in Python using exact integer arithmetic. The accompanying public GitHub repository contains four scripts: an exploratory computation up to 10^4 , a modular refinement analysis up to 10^5 , a large-scale profile analysis up to 10^6 , and a final streaming computation up to 10^7 . The final script does not store full trajectories; instead, it updates resistance-class counts, retained top temporal-resistance cases, retained top amplitude-resistance cases, modular residue summaries, and modular resistance-spine statistics in streaming form.

In the output files, the column label P denotes the same relative peak ratio written as $\rho(n)$ in this paper. The Python scripts used to generate the reported numerical results are publicly available at <https://github.com/seymus1/collatz-pre-descent-resistance>.

Author Contributions

Seyma Yaman Kayadibi is the sole author of this work and was responsible for the conceptualization, mathematical formulation, computational implementation, analysis, interpretation of results, and manuscript preparation.

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Not applicable.

Consent to Participate

Not applicable.

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